Palo Alto Networks Cloud Security Engineer

Datasheet

JULY 2026

The Palo Alto Networks Cloud Security Engineer certification validates the knowledge, skills, and abilities required of experienced cloud security engineers in planning for Cloud-Native Application Protection Platform (CNAPP) deployments, cloud account and data source onboarding, security posture management across different domains, cloud workload protection (CWP), Cloud Detection and Response (CDR), application security workflows, troubleshooting common platform issues, and automated remediation. It confirms the ability to maintain end-to-end traceability of issues and implement security guardrails throughout the software development lifecycle (SDLC).

The purpose of this document is to help you prepare for the exam and attain the certification. Please note that this document is intended to help identify the topics covered and to provide resources and references for understanding those topics. It is not intended to be used as the sole document to prepare for the Cloud Security Engineer exam.

Exam Details:

- Duration: 90 minutes
- Format: Multiple-choice questions
- Language: English
- Cost: $250 USD*
- Delivered by: Pearson Professional Assessments

** Price may vary by country*

Audience and Qualifications

Target Audience

This exam is designed for security engineers, professional services consultants, DevSecOps engineers, technical support engineers, customer success engineers, and security operations engineers. It is also designed for individuals responsible for deploying and managing cloud security tooling and ensuring compliance across integrated cloud platforms. It is intended for individuals with 3+ years in a cloud security-related field, and 1–2 years with Palo Alto Networks cloud security solutions or Cortex platform and/or with other Cloud-Native Application Protection Platform (CNAPP) solutions.

Skills Required

- Working knowledge of network security
- Working knowledge of TCP/IP and how traffic is directed within a network, including routing protocols
- Working knowledge of networking infrastructure, protocols, and topology
- Basic understanding of endpoint operating system fundamentals and security hardening methods
- Working knowledge of cloud security and security automation technology
- Basic knowledge of current and emergent trends in information security
- Understanding of the use of security models/architectures (e.g., defense-in-depth, Zero Trust)
- Engineering-level knowledge of Cortex Cloud
- Working knowledge of microservices, containerization (Docker), and orchestration (Kubernetes) design patterns
- Deep understanding of the security boundaries between the cloud service provider (CSP) and the customer
- Ability to read, interpret, and suggest security improvements for Terraform, CloudFormation, Bicep, and Helm charts
- Understanding of modern DevOps workflows and where security "guardrails" (e.g., SCA, SAST, IaC scanning) are most effectively placed
- Ability to work with JSON and YAML payloads for policy definitions, API interactions, and log analysis
- Working knowledge of SAML 2.0, OIDC, and OAuth for both user access and machine-to-machine integrations
- Skill in crafting granular IAM policies that balance functionality with security across multicloud environments
- Understanding of how Cortex Cloud controls map to industry standards such as NIST SP 800-53, SOC2, HIPAA, and PCI-DSS
- Knowledge of the CVSS scoring system and the ability to prioritize remediation based on environmental context.

- Experience using AWS CLI, Azure CLI, GCloud CLI, and kubectl
- Ability to navigate the Linux filesystem and manage services
- Experience using tools such as Postman or cURL to interact with REST APIs for automation or data retrieval

Blueprint

The blueprint table lists the domains covered and includes domain weighting. The percentage weights represent the portion of the exam score that is attributed to each domain. Many candidates find the table provides focus for studies during exam preparation. Also included in the blueprint table are the more specific tasks associated with each domain. Pay particular attention to these tasks, as they provide more targeted areas of study within the domains.

1. Planning and Installation 12%

1.1 Evaluate the existing cloud and IT infrastructure to align with Cortex Cloud architecture (i.e., multicloud footprint, workload inventory, Cloud-Native Application Protection Platform [CNAPP] tooling, migration)

1.2 Determine Cortex Cloud deployment and license requirements, data source ingestion and retention targets, scanning models (e.g., cloud scan versus Outpost), and cloud provider identity and access management (IAM) requirements (e.g., least-privilege)

1.3 Determine network communication and connectivity requirements for Cortex Cloud components (e.g., egress rules, URLs, ports, regional endpoints)

1.4 Explain the configuration and management lifecycle of user identity, roles, permissions, and access controls within a Cortex tenant (e.g., Scope-Based Access Control [SBAC], custom Role-Based Access Control [RBAC] roles, API keys)

1.5 Identify and describe cost optimization methods (e.g., scan cadence, duplicate audit logs)

2. Integration 16%

2.1 Explain the onboarding and configuration process for cloud provider resources (e.g., AWS, Azure, GCP, OCI)

- 2.1.1 Security capabilities (e.g., Data Security Posture Management [DSPM], audit logs, Cortex XSIAM analytics, agentless scanning, container registry options, serverless function scanning)
- 2.1.2 Cloud scan versus Outpost

2.2 Explain the process of onboarding and configuring data and AI sources (e.g., Office 365, Databricks, Azure Foundry)

2.3 Explain the process of onboarding application development and deployment systems (e.g., GitHub, Jenkins)

2.4 Explain the process of container registry integration (e.g., JFrog, Docker, GitLab)

2.5 Explain the integration process for third-party static application security testing (SAST) / Software Composition Analysis (SCA) tools (e.g., Semgrep, Veracode, Generic SARIF)

2.6 Explain the deployment and configuration process for Broker VM

3. Posture Security 22%

- **3.1 Explain the process of dashboard creation and configuration using XQL**
- **3.2 Explain the creation and configuration process for static and dynamic asset groups**
- **3.3 Explain the creation and configuration of cloud security rules and policies (e.g., attack path, configuration, data, identity, network exposure, AI)**
- **3.4 Explain the creation and configuration of cloud workload rules and policies (e.g., custom rules, scanner type, misconfiguration, malware, secrets, trusted images)**
- **3.5 Explain the creation and management of vulnerability policies (i.e., issue creation and prevention)**
- **3.6 Identify and describe methods for managing custom and out-of-the-box compliance standards / controls, assessments, and reporting**
- **3.7 Explain the deployment of Kubernetes connectors**
- **3.8 Identify and describe data security classification settings and scanning options**

4. Runtime Security 18%

- **4.1 Explain the deployment and configuration of XDR agents (e.g., virtual machines VMs, Containers as a Service [CaaS], serverless functions, Kubernetes)**
- **4.2 Explain the process of endpoint protection configuration**
  - 4.2.1 Endpoint profiles and policies (e.g., prevention, extension, exceptions)
  - 4.2.2 Endpoint groups
- **4.3 Explain the process of third-party API gateway integration (e.g., AWS, API Gateway, APIM, Apigee)**
- **4.4 Explain the creation and configuration process for Cloud Detection and Response (CDR) and Threat Management**
  - 4.4.1 Detection rules (e.g., correlation, behavioral indicator of compromise [BIOC], indicator of compromise [IOC], analytic)
  - 4.4.2 Automation playbooks

5. Application Security 16%

- **5.1 Identify and evaluate Application Security Posture Management (ASPM) features and functionality**
  - 5.1.1 Business applications (i.e., manual, dynamic)
  - 5.1.2 Service-level agreement (SLA) for issues
- **5.2 Explain rule and policy deployment and management**
  - 5.2.1 Code security
  - 5.2.2 AI-recommended guardrails
  - 5.2.3 CI/CD policies
- **5.3 Explain the configuration and management of repository scanning (e.g., Infrastructure as Code [IaC], secrets, SCA, IaC drift detection)**
- **5.4 Explain the deployment and configuration of Cortex Cloud application security IDE plugins (i.e., VS Code, JetBrains)**
- **5.5 Identify and describe the steps to install, configure, and execute Cortex CLI in workflows**

6. Troubleshooting 16%

- **6.1 Identify and describe troubleshooting issues related to insufficient cloud service provider (CSP) IAM permissions, CloudFormation / Terraform stack failures, and organization-level onboarding related errors, including ingestion**
- **6.2 Demonstrate knowledge of troubleshooting Cortex Cloud Workload components (e.g., agents / connectors, connectivity issues)**
- **6.3 Demonstrate knowledge of troubleshooting communication gaps between customer-managed outposts and the Cortex tenant**
- **6.4 Demonstrate knowledge of authentication failure resolution (OAuth / PAT) and webhook delivery issues when connecting version control systems (VCS) (e.g., GitHub, GitLab) and CI/CD systems (e.g., Jenkins) to Cortex Cloud**
- **6.5 Identify and describe troubleshooting issues related to Application Security components (e.g., proper policy configuration)**
- **6.6 Identify and describe access and permissions troubleshooting issues**
- **6.7 Explain the process of diagnosing / determining the source of an unexpected Cloud cost increase**

English as a Second Language (ESL) Accommodation

All exams are delivered worldwide in English. A 30-minute time extension is provided by default to candidates testing in non-English speaking countries.

Learning Resources

Learning Path

Related Palo Alto Networks training resources can be found here.

Recommended Preparation

Palo Alto Networks certification exam items are developed and approved by exam development experts in conjunction with subject matter experts (SMEs) who represent a broad spectrum of roles relevant to each certification. Each item is referenced to a publicly available technical or scholarly source.

This datasheet provides the full blueprint to which the exam items are written. A solid foundation of the knowledge, skills, and abilities required to pass this certification exam can be achieved by researching keywords from each topic area included in the blueprint—either through the use of an AI-powered browser or directly within Palo Alto Networks documentation—and completing relevant coursework found in the learning path.

Exam Registration

Register for Palo Alto Networks certification exams at Pearson Professional Assessments here.

paloalto NETWORKS

3000 Tannery Way
Santa Clara, CA 95054

Main: +1.408.753.4000
Sales: +1.866.320.4788
Support: +1.866.898.9087

www.paloaltonetworks.com

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